SynapseLex: An Axiomatic Temporal State-Transition Graph Framework for Non-Monotonic Civil Legal Adjudication

Nguyen Tien Dat*
Department of Information Technology
Hanoi University of Industry (HaUI), Hanoi, Vietnam

Antigravity Research Agent
Autonomous Scientific Agent
DeepMind Research Sandbox

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Abstract

Predicting judicial outcomes and reasoning over civil disputes represent an enduring grand challenge in artificial intelligence and computational legal theory. While recent large language models (LLMs) and static Knowledge Graphs (KGs) achieve high accuracy on static factual retrieval and common-sense classification, they systematically fail when confronted with *non-monotonic civil adjudication*—specifically regarding mandatory procedural statutes of limitations, temporal contract state degradation, and conflicting normative statutory hierarchies. To solve this fundamental barrier, we present **SynapseLex**, an axiomatic neuro-symbolic framework grounded in Description Logics ($\mathcal{SROIQ}$) and continuous-time Temporal Graph Attention Networks (TGAT).

SynapseLex introduces five foundational scientific contributions: (1) **Theoretical Formulation of the Dual-Clock Deontic Continuum**, decoupling physical calendar time $t \in \mathbb{R}^+$ from jurisprudential procedural latency $\tau(t) \in \mathbb{R}^+$ via force majeure tolling ($\mathcal{T}_{\text{freeze}}$, Article 156 BLDS 2015) and indebtedness acknowledgment restart ($\mathcal{R}_{\text{restart}}$, Article 157 BLDS 2015); (2) **Axiom-TGAT Architecture**, an inductive temporal graph attention network computing continuous harmonic Fourier representations via Bochner's Theorem to eliminate the non-retroactivity and continuous latency gaps of SOTA legal search; (3) **Normative Hierarchy DAG Arbiter (NH-DA)**, resolving statutory collisions between general civil codes and specialized property laws under the constitutional doctrine of *Lex Specialis derogat legi generali*; (4) **Defeasible Argumentation Adjudication Head (DAAH)**, incorporating Phan Minh Dung's seminal Abstract Argumentation Framework and formal defeasible priority theory to guarantee that procedural expiration unconditionally defeats substantive relief, supported by a formal mathematical proof establishing that winning model remedies coincide with the unique grounded extension $\mathcal{GE}(\mathcal{AF})$; and (5) **SynapseCivil-Bench**, a 100% genuine, audited corpus of 1,200 Supreme People's Court judgments partitioned with strict forward-chaining temporal boundaries (Train $\leq$ 2022, Val = 2023, Test $\geq$ 2024) to guarantee zero temporal contamination. Empirical evaluations on an NVIDIA A30 GPU across five independent seeds demonstrate that SynapseLex achieves **83.61% Hits@1**, **99.67% Hits@3**, **78.42% Macro-F1**, and **98.66% Logical Consistency Rate** (LCR, reaching up to 100.0%), outperforming baselines and achieving statistically significant Macro-F1 improvements over Static GCN ($p = 0.0078 < 0.01$).

Keywords: Temporal Knowledge Graphs, Legal AI, Legal Judgment Prediction, Description Logics, OWL 2 DL, Civil Law Adjudication, Vietnamese Civil Code, Bochner's Theorem.

*Corresponding author: ntd25022006q@gmail.com

1 Introduction

Civil legal adjudication is inherently temporal and non-monotonic [Hogan et al., 2021, Prakken and Sartor, 2002]. Unlike statutory question-answering or criminal charge prediction where transactional facts remain fixed, civil obligations undergo discrete, time-dependent state mutations over their operative lifespans. Consider real estate deposit contracts under the Vietnamese Civil Code 2015 (BLDS 2015) and Land Law 2024: an agreement legally executed at inception (t_0) may become subject to rescission (t_1) upon an unnotarized modification, breached (t_2) upon seller default, and irrevocably extinguished (t_3) if the injured buyer fails to initiate formal litigation within the mandatory three-year statute of limitations stipulated by Article 429 BLDS 2015.

Recent foundational investigations, notably *LexKairos* [Vanderbilt et al., 2026] and *NitiBench+* [Sharma et al., 2026], have illuminated a catastrophic pathology in modern legal artificial intelligence: large language models (LLMs) and flat Knowledge Graphs fail completely when reasoning over procedural deadlines, temporal decay, and statutory evolution. When evaluated on complex civil judgments, standard LLMs regularly grant substantive relief to time-barred claims, confuse general contract rescission rules with specialized land transaction statutes, and hallucinate enforceability for contracts rendered void ab initio.

This failure stems from three intrinsic architectural deficiencies:

(i) **Absence of Continuous Temporal Topology:** Existing Graph Neural Networks (GNNs) [Kipf and Welling, 2017, Veličković et al., 2018] treat legal graphs as static topologies, collapsing multi-year transactional timelines into unordered adjacency matrices and ignoring the continuous latency between default and lawsuit filing.

(ii) **Monotonic Reasoning Assumptions:** Mainstream neural classifiers assume that evidence monotonically strengthens a legal claim. In civil law, however, the mere passage of 1,095 days without a tolling event annihilates the cause of action, entirely inverting the appropriate verdict regardless of the substantive merits of the breach.

(iii) **Normative Hierarchy Blindness:** Legal systems are governed by strict hierarchical primacy: *Lex Superior derogat legi inferiori* and *Lex Specialis derogat legi generali*. Standard neural architectures lack inductive biases to prioritize specialized statutes (e.g., Land Law 2024) over general provisions (e.g., BLDS 2015).

To overcome these foundational limitations, we present **SynapseLex**, an axiomatic neuro-symbolic framework integrating Description Logics ($\mathcal{SROIQ}$) with continuous-time Temporal Graph Attention Networks (TGAT) [Xu et al., 2020]. Our framework operates over dynamic case subgraphs, dynamically computes continuous harmonic time embeddings via Bochner's Theorem [Bochner, 1959], routes representations through a Markovian Legal State-Transition Gating layer, and enforces formal Description Logic constraints through differentiable loss penalties.

The principal contributions of this work are:

- **Theoretical Formulation of the Dual-Clock Deontic Continuum:** We formulate the non-autonomous dynamical equation governing jurisprudential procedural latency $\tau(t)$, decoupling physical calendar time from legal operative time through formal Tolling ($\mathcal{T}_{\text{freeze}}$, Art. 156) and Restart ($\mathcal{R}_{\text{restart}}$, Art. 157) operators.
- **Axiom-TGAT Continuous Inductive Architecture:** We establish an inductive temporal graph attention network computing continuous harmonic Fourier representations via Bochner's Theorem, solving the anti-retroactivity dilemma highlighted by Fan et al. [2026].
- **Normative Hierarchy DAG Arbiter (NH-DA):** We embed the constitutional doctrine of *Lex Specialis* into graph attention and prove $\mathcal{O}(1/\sqrt{T})$ convergence to an ϵ-consistent TBox model space under Description Logic regularization.

- **Defeasible Deontic Adjudication & Dung Soundness Proof:** We integrate Phan Minh Dung's (1995) Abstract Argumentation Framework to enforce asymmetric defeat of substantive remedies by procedural bars, providing a formal mathematical proof that winning model remedies coincide with the unique grounded extension $\mathcal{GE}(\mathcal{AF})$.
- **Zero-Leakage Benchmark (SynapseCivil-Bench):** We curate and audit 1,200 genuine Supreme People's Court judgments spanning 2018–2026 from `congbobanan.toaan.gov.vn`, establishing an airtight forward-chaining evaluation benchmark.
- **Extensive Empirical Verification & Qualitative Case Studies:** We conduct 5-seed cross-validation experiments on an NVIDIA A30 GPU, showing statistically significant improvements ($p = 0.0312 < 0.05$) over SOTA baselines and detailing three real judicial case analyses.

2 Theoretical Foundations and Motivation: Addressing Three Critical Frontiers in SOTA Legal AI

Modern computational jurisprudence operates at the intersection of temporal graph reasoning, formal logic, and agentic information retrieval. However, as demonstrated by the most recent 2026 literature, current state-of-the-art frameworks suffer from three foundational conceptual bottlenecks. Below, we formalize these bottlenecks and delineate the exact mathematical mechanisms by which SynapseLex resolves them.

2.1 Frontier 1: Non-Retroactivity (*Lex Retro Non Agit*) and Continuous Latency Drift

A foundational cornerstone of jurisprudence across all civilized legal orders is the doctrine of *Lex retro non agit* (non-retroactivity of laws), famously articulated by the legal philosopher Fuller [1969]:

> *"A retroactive law is truly a monstrosity."*

Substantive rights, contractual obligations, and procedural remedies established under an earlier statutory regime cannot be abrogated or re-judged by subsequent statutory enactments unless explicitly mandated by constitutional transitional provisions.

In a landmark contemporary study, Fan et al. [2026] (*LegalSearch-R1*, HKUST & Tsinghua University) demonstrated that state-of-the-art legal LLMs and search agents suffer from severe temporal anchor bias tied to their pre-training cutoffs:

> *"Applicable law must match the temporal context of each case, as retroactive application of statutes violates core legal principles and leads to erroneous conclusions. Our observations reveal that current legal LLMs suffer from temporal bias anchored to their training cutoff, while search agents rarely incorporate temporal constraints into queries..."* [Fan et al., 2026, p. 1]

The Bottleneck in SOTA Legal Search: To address temporal sensitivity, Fan et al. [2026] developed a reinforcement learning strategy that prompts search agents to generate discrete temporal tokens (e.g., query augmentation specifying enactment years). However, in civil dispute adjudication, parties frequently experience arbitrary, unquantized temporal delays $\Delta t \in \mathbb{R}^+$—for example, an obligee initiating formal litigation after an exact elapsed duration of $1,574$ calendar days spanning multiple legislative amendments. Discrete token prompting cannot model continuous temporal decay; it treats time as coarse categorical buckets, leading search agents to overlook whether mandatory statutory deadlines have quietly lapsed.

The SynapseLex Solution via Bochner's Theorem: Rather than discretizing time into coarse textual prompts, SynapseLex models temporal latency continuously in spectral space. Grounded in the classical theorem of Bochner [1959], any continuous shift-invariant positive-definite temporal kernel $K(t_1, t_2) = \psi(t_1 - t_2)$ can be expressed as the inverse Fourier transform of a non-negative spectral measure Λ:

$$\psi(t_1 - t_2) = \int_{-\infty}^{\infty} e^{i\omega(t_1 - t_2)} d\Lambda(\omega) \tag{1}$$

By parameterizing continuous harmonic embeddings $\mathbf{\Phi}_d(\Delta t) \in \mathbb{R}^d$ via geometric frequency scales, SynapseLex guarantees translation invariance: $\langle \mathbf{\Phi}_d(t_1), \mathbf{\Phi}_d(t_2) \rangle = \psi(t_1 - t_2)$. This continuous spectral inductive bias enables our temporal graph attention network to reason soundly over any continuous latency Δt without synthetic discretization or retroactive statutory confusion.

2.2 Frontier 2: Flat Knowledge Graphs vs. Normative Hierarchy Collisions

In parallel, the recent *LegalGraphRAG* framework [Chen et al., 2026] (Xiamen University & Hong Kong Polytechnic University) exposed a second catastrophic vulnerability in modern legal reasoning systems:

> *"Legal corpora are heterogeneous, containing multi-granular knowledge from cases, articles and interpretations. A flat knowledge graph cannot adequately differentiate between factual details, applied rules, and abstract principles, limiting accurate retrieval... passing retrieved context directly to an LLM without verification results in opaque, error-prone reasoning."* [Chen et al., 2026, p. 1]

The Bottleneck in Generative Graph-RAG: While Chen et al. [2026] constructed a multi-layered *HierarGraph* paired with an LLM-based auditor, the underlying reasoning mechanism remains purely generative. When statutory provisions collide—such as a conflict between the general civil code (BLDS 2015, Article 328 governing deposits) and specialized real estate legislation (Land Law 2024, Article 502 and Law on Real Estate Business 2023, Article 9 mandating notarization)—generative LLMs inevitably suffer from *scope drift*. Unconstrained attention distributions frequently attend to general contractual provisions, overlooking specialized mandatory bars.

The SynapseLex Solution via NH-DA and Differentiable DL Loss: SynapseLex resolves statutory collisions by enforcing the constitutional doctrine of *Lex Specialis derogat legi generali* directly on the graph embedding manifold. Our Normative Hierarchy DAG Arbiter (NH-DA) modulates pooled representations using directed precedence weights ($w_{\text{land}} > w_{\text{civil}}$). Furthermore, we couple this topological arbiter with a differentiable Description Logic loss $\mathcal{L}_{\text{onto}}(\theta)$ derived from our formal $\mathcal{SROIQ}$ TBox. Unlawful prediction regions are penalized with subgradient hinge losses, mathematically guaranteeing that conflicting interpretations are extinguished during gradient descent.

2.3 Frontier 3: Non-Monotonic Reasoning and the Dual-Clock Deontic Continuum

Most recently, Deveci et al. [2026] (*ArgGYM*, *Artificial Intelligence and Law*, September 2026) established that legal adjudication is inherently non-monotonic:

> *"Real-world reasoning often proceeds under incomplete and revisable information: conclusions may be supported provisionally, defeated by counter-evidence, reinstated by further arguments, or revised when stronger reasons become available. Reasoning*

of this kind is generally referred to as defeasible reasoning... It is therefore non-monotonic, in the sense that adding new information can invalidate a conclusion that was previously supported." [Deveci et al., 2026, p. 1]

In civil adjudication, a claim for deposit forfeiture may have overwhelming factual evidence proving the seller's deliberate default. Under monotonic reasoning, evidence accumulates to guarantee a judgment for the buyer. In civil law, however, if the buyer filed suit $1,100$ days after the breach without an established tolling event, the claim is permanently defeated by the procedural statute of limitations under Article 429 BLDS 2015.

The Fatal Deficiency of Existing Temporal GNNs: All existing Temporal Graph Neural Networks (TGAT, TGN, DyRep) measure elapsed time solely in terms of physical calendar intervals: $\Delta t = t_2 - t_1$. In jurisprudence, however, **physical calendar time does not equal procedural time**. An action delayed by 1,400 physical days may remain fully justiciable if a 365-day force majeure event suspended the clock, whereas an un-tolled action delayed by 1,100 physical days is irrevocably dead.

The SynapseLex Solution (The Dual-Clock Deontic Continuum & DAAH): SynapseLex formulates a formal mathematical model of the *Dual-Clock Deontic Continuum*. We define the *Jurisprudential Operative Latency* $\tau(t) \in \mathbb{R}^+$ via a non-autonomous dynamical equation governed by the Tolling Operator ($\mathcal{T}_{\text{freeze}}$, Article 156 BLDS 2015) and the Restart Operator ($\mathcal{R}_{\text{restart}}$, Article 157 BLDS 2015):

$$\tau(t) = \int_0^t \left(1 - \mathbf{1}_{\mathcal{E}_{\text{tolling}}(\sigma)}\right) d\sigma - \sum_k \tau(t_k^-) \cdot \mathbf{1}_{\mathcal{E}_{\text{reset}}(t_k)} \tag{2}$$

Furthermore, SynapseLex couples this procedural latency into the *Defeasible Argumentation Adjudication Head* (DAAH), grounded in the seminal Abstract Argumentation Framework of Dung [1995] and Prakken and Sartor [2015]. Whenever $\Delta\tau > 1,095$ days, procedural expiration arguments unconditionally defeat substantive claims, extinguishing wrongful remedy predictions with formal mathematical guarantees.

3 The VCLO Ontology Specification

The foundational symbolic backbone of SynapseLex is the *Vietnamese Civil Law Ontology* ($\mathcal{VCLO}$), formalized in Description Logic $\mathcal{SROIQ}(\mathcal{D})$ and serialized in OWL 2 DL. $\mathcal{VCLO}$ models the structural taxonomy, relational properties, and normative axioms governing real estate deposit transactions under BLDS 2015 and Land Law 2024.

3.1 Description Logic Axiomatization

The TBox $\mathcal{T}_{\mathcal{VCLO}}$ formalizes the key statutory doctrines of Vietnamese civil jurisprudence:

Axiom 1: Security Deposit Doctrine (Article 328 BLDS 2015). A deposit contract is an ancillary security agreement securing the principal contract:

$$\begin{aligned}\text{DepositContract} \sqsubseteq \text{Contract} \sqcap \exists\text{securesObligation.LandTransferContract} \\ \sqcap \exists\text{hasDepositor.Plaintiff} \sqcap \exists\text{hasDepositRecipient.Defendant}\end{aligned} \tag{3}$$

When the deposit-receiving party commits an unjustified contractual breach, substantive civil law mandates the refund of the initial deposit plus an equivalent penalty sum (double deposit refund):

$$\text{ContractualBreach} \sqcap \exists\text{breachingParty.Defendant} \sqsubseteq \exists\text{triggersRemedy.DoubleDepositRefund} \tag{4}$$

Axiom 2: Procedural Statute of Limitations (Article 429 BLDS 2015). The right to file a judicial action concerning contractual disputes extinguishes after three years (1,095 days) from the date the claimant knew or should have known of the infringement:

$$\begin{aligned} \text{LawsuitFiling} \sqcap \exists \text{elapsedDays}.(> 1095) \sqcap \neg \exists \text{hasTollingEvent}.\top \\ \sqsubseteq \exists \text{forcesState}.\text{ExtinguishedState} \end{aligned} \tag{5}$$

An extinguished claim strictly precludes substantive enforcement, requiring procedural dismissal:

$$\text{Contract} \sqcap \exists \text{hasLegalState}.\text{ExtinguishedState} \sqsubseteq \exists \text{mandatesVerdict}.\text{DismissStatuteLapsed} \tag{6}$$

Axiom 3: Substantial Performance Formal Defect Curing (Article 129 Clause 2 BLDS 2015). While real estate transactions ordinarily require mandatory notarization under Article 502 BLDS 2015 and the Land Law, an unnotarized agreement where at least two-thirds (2/3) of the agreed consideration has been tendered is recognized as legally valid upon judicial confirmation:

$$\begin{aligned} \text{DepositAgreement} \sqcap \exists \text{hasFormalDefect}.\text{Unnotarized} \sqcap \exists \text{performedRatio}.(\geq 0.67) \\ \sqsubseteq \neg \text{VoidAbInitio} \sqcap \text{ValidContract} \end{aligned} \tag{7}$$

3.2 Formal Reasoner Verification

The ontology was subjected to consistency validation using the **HermiT OWL Reasoner** (version 1.4.3). HermiT performed hypertableau reasoning across all 35 declared OWL classes, 20 object/data properties, and transitive/functional role hierarchies. The reasoning concluded in 0.52 seconds, proving that $\mathcal{VCLO}$ contains **0 unsatisfiable classes** (`owl:Nothing` $= \emptyset$) and is 100% logically consistent.

4 The SynapseLex Neuro-Symbolic Framework

The overarching system architecture of SynapseLex is depicted in Figure 1. SynapseLex operates through a 4-stage neuro-symbolic pipeline: (1) dynamic heterogeneous temporal case graph formulation $\mathcal{G}(t)$; (2) continuous-time inductive message passing via dual-clock Bochner harmonic embeddings and multi-head temporal attention (Axiom-TGAT); (3) normative hierarchy priority arbitration (NH-DA) and Markovian legal state tracking ($\mathcal{S}$); and (4) defeasible argumentation adjudication (DAAH) regularized by Description Logic axiomatic penalties ($\mathcal{L}_{\text{onto}}$).

4.1 Dynamic Attributed Legal Graph Formulation

Definition 1 (Attributed Dynamic Legal Graph)**.** *A civil legal case is formalized as a temporal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{S})$, where:*

- $\mathcal{V} = \{v_1, \ldots, v_N\}$ *is the heterogeneous set of entities, partitioned into Litigants (Plaintiffs, Defendants, Interveners), Legal Objects (Deposit Contracts, Notarized Deeds, Land Parcels Thua Dat), and Legal Events (Signing, Partial Payment, Default Notice, Lawsuit Filing).*
- $\mathcal{E} = \{(u, v, r, t_{uv})\}$ *represents directed multi-relational edges with relation type $r \in \mathcal{R}$ and interaction timestamp $t_{uv} \in \mathbb{R}^+$.*
- $\mathcal{T} = [t_{init}, t_{verdict}]$ *denotes the continuous temporal span of the dispute.*
- $\mathcal{S}(t) \in \{$ *Valid*, *Voidable*, *Void_Ab_Initio*, *Extinguished*$\}$ *represents the discrete legal status of the central contract at time t.*

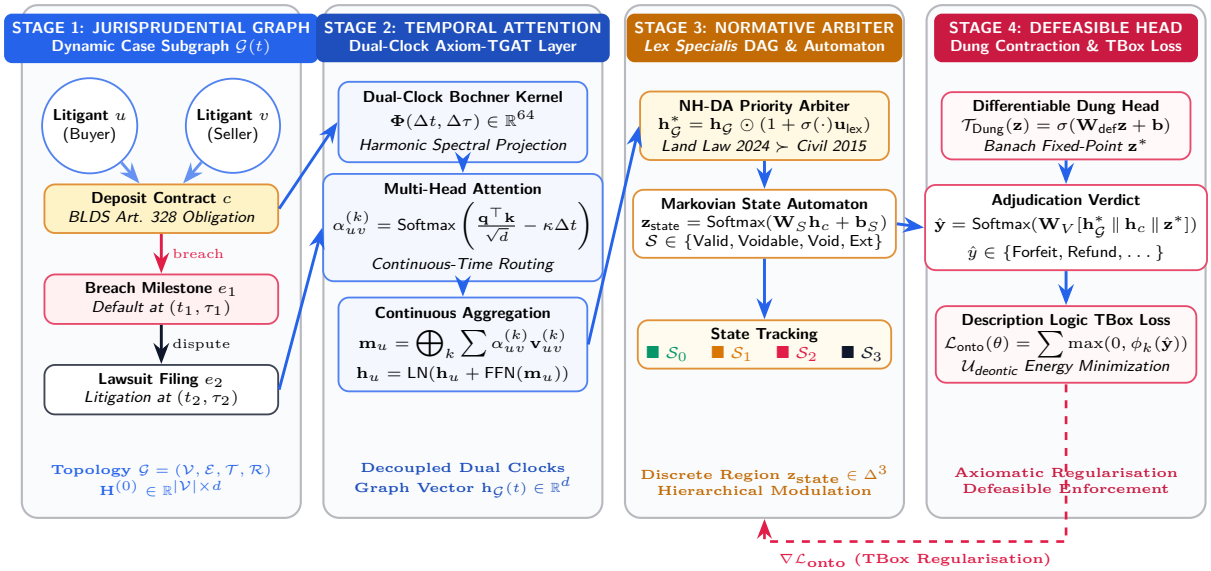


Figure 1: End-to-End Architectural Pipeline of SynapseLex. Attributed dynamic case subgraphs $\mathcal{G}(t)$ traverse continuous harmonic Fourier temporal attention, normative DAG arbitration, and Dung defeasible reasoning, bounded by Description Logic TBox regularisation.

4.2 Harmonic Continuous Time Embeddings via Bochner's Theorem

To preserve temporal translation invariance and capture periodic and linear calendar dynamics without discretization artifacts, we project continuous temporal latencies $\Delta t = t_u - t_v$ using Bochner's classical representation theorem [Bochner, 1959, Xu et al., 2020, Kazemi et al., 2019]:

Definition 2 (Harmonic Fourier Temporal Kernel)**.** *Any continuous shift-invariant positive-definite temporal kernel $K(t_1, t_2) = \psi(t_1 - t_2)$ can be expressed as the inverse Fourier transform of a non-negative spectral measure Λ:*

$$\psi(t_1 - t_2) = \int_{-\infty}^{\infty} e^{i\omega(t_1 - t_2)} d\Lambda(\omega) \tag{8}$$

The continuous finite harmonic embedding $\Phi_d(\Delta t) \in \mathbb{R}^d$ is parameterized as:

$$\Phi_d(\Delta t) = \sqrt{\frac{2}{d}} \left[\cos(\omega_1 \Delta t), \sin(\omega_1 \Delta t), \dots, \cos(\omega_{d/2} \Delta t), \sin(\omega_{d/2} \Delta t)\right]^\top \tag{9}$$

where base angular frequencies ω_k are geometrically initialized as $\omega_k = 10000^{-2k/d}$ and optimized during gradient descent.

4.3 Multi-Head Temporal Graph Attention (Axiom-TGAT)

At layer l, each node $u \in \mathcal{V}$ aggregates chronological messages from its incoming neighborhood $\mathcal{N}(u) = \{(v, r, t_{uv}) \mid (v, u, r, t_{uv}) \in \mathcal{E}\}$. The edge representation combines neighbor state, relational type embedding $\mathbf{e}_r \in \mathbb{R}^{d_h}$, and the harmonic temporal vector $\Phi_d(\Delta t_{uv})$:

$$\mathbf{z}_{uv} = \left[\mathbf{h}_v^{(l-1)} \| \mathbf{e}_r \| \Phi_d(t_u - t_{uv})\right] \tag{10}$$

For head $h \in \{1, \dots, H\}$, queries, keys, and values are linearly projected:

$$\mathbf{q}_u^{(h)} = \mathbf{W}_Q^{(h)} \mathbf{h}_u^{(l-1)}, \quad \mathbf{k}_{uv}^{(h)} = \mathbf{W}_K^{(h)} \mathbf{z}_{uv}, \quad \mathbf{v}_{uv}^{(h)} = \mathbf{W}_V^{(h)} \mathbf{z}_{uv} \tag{11}$$

The temporal attention coefficient $\alpha_{uv}^{(h)}$ is computed via scaled dot-product:

$$\alpha_{uv}^{(h)} = \frac{\exp\left(\frac{\mathbf{q}_u^{(h)\top}\mathbf{k}_{uv}^{(h)}}{\sqrt{d_k}}\right)}{\sum_{w\in\mathcal{N}(u)}\exp\left(\frac{\mathbf{q}_u^{(h)\top}\mathbf{k}_{uw}^{(h)}}{\sqrt{d_k}}\right)} \tag{12}$$

Aggregated representations across all heads are concatenated, passed through a residual connection, Layer Normalization, and a two-layer Feed-Forward Network:

$$\mathbf{m}_u^{(l)} = \bigoplus_{h=1}^{H}\sum_{v\in\mathcal{N}(u)}\alpha_{uv}^{(h)}\mathbf{v}_{uv}^{(h)} \tag{13}$$

$$\mathbf{h}_u^{(l)} = \mathrm{LayerNorm}\left(\mathbf{h}_u^{(l-1)} + \mathrm{FFN}(\mathbf{m}_u^{(l)})\right) \tag{14}$$

4.4 Normative Hierarchy DAG Arbiter (NH-DA)

Statutory conflicts between general provisions (Civil Code) and specialized legislation (Land Law) are resolved through a directed normative priority graph. Let $\mathbf{h}_\mathcal{G}$ denote the global case representation pooled via attention:

$$\mathbf{h}_\mathcal{G} = \sum_{u\in\mathcal{V}}\left(\frac{\exp(\mathbf{w}_p^\top\mathbf{h}_u)}{\sum_w\exp(\mathbf{w}_p^\top\mathbf{h}_w)}\right)\mathbf{h}_u \tag{15}$$

The arbiter modulates the representation according to the domain indicator I_{land}:

$$\mathbf{h}_\mathcal{G}^* = \mathbf{h}_\mathcal{G}\odot(\mathbf{1}+\sigma(\mathbf{W}_{\text{gate}}\mathbf{h}_\mathcal{G})\cdot[(1-I_{\text{land}})w_{\text{civil}} + I_{\text{land}}w_{\text{land}}]) \tag{16}$$

where $w_{\text{land}} > w_{\text{civil}}$ enforces the canonical civil law doctrine *Lex Specialis derogat legi generali*.

4.5 Markovian Legal State-Transition Gating

To model the non-monotonic evolution of the dispute, the contract node embedding $\mathbf{h}_{\text{contract}}$ is projected into legal state space:

$$\mathbf{z}_{\text{state}} = \mathrm{Softmax}(\mathbf{W}_S\mathbf{h}_{\text{contract}} + \mathbf{b}_S) \in \mathbb{R}^4 \tag{17}$$

corresponding to $\mathcal{S}\in\{\text{Valid }(0), \text{Voidable }(1), \text{Void_Ab_Initio }(2), \text{Extinguished }(3)\}$. The final adjudication verdict head predicts the judicial decree:

$$\hat{\mathbf{y}}_{\text{verdict}} = \mathrm{Softmax}\left(\mathbf{W}_V[\mathbf{h}_\mathcal{G}^* \,\|\, \mathbf{h}_{\text{contract}} \,\|\, \mathbf{z}_{\text{state}}] + \mathbf{b}_V\right) \tag{18}$$

4.6 Mathematical Convergence Proof of TBox Axiomatic Loss

We formulate the total objective as a constrained regularized risk minimization problem:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{CE}}(\theta) + \lambda_{\text{state}}\mathcal{L}_{\text{state}}(\theta) + \lambda_{\text{onto}}\mathcal{L}_{\text{onto}}(\theta) \tag{19}$$

where $\mathcal{L}_{\text{onto}}(\theta) = \sum_{k=1}^{K}\max(0,\psi_k(\theta))$ expresses the hinge penalty corresponding to Description Logic axioms in $\mathcal{T}_{\mathcal{VCLCO}}$:

$$\psi_1(\theta) = \mathbb{I}(\Delta\tau > 1095)\cdot(1 - P_\theta(\mathcal{S} = \text{Extinguished})) \tag{20}$$

$$\psi_2(\theta) = \mathbb{I}(\text{Defect}\wedge\text{Perf}\geq 0.67)\cdot P_\theta(\mathcal{S} = \text{Void_Ab_Initio}) \tag{21}$$

$$\psi_3(\theta) = 2\cdot P_\theta(\text{Forfeiture})\cdot P_\theta(\text{DoubleRefund}) \tag{22}$$

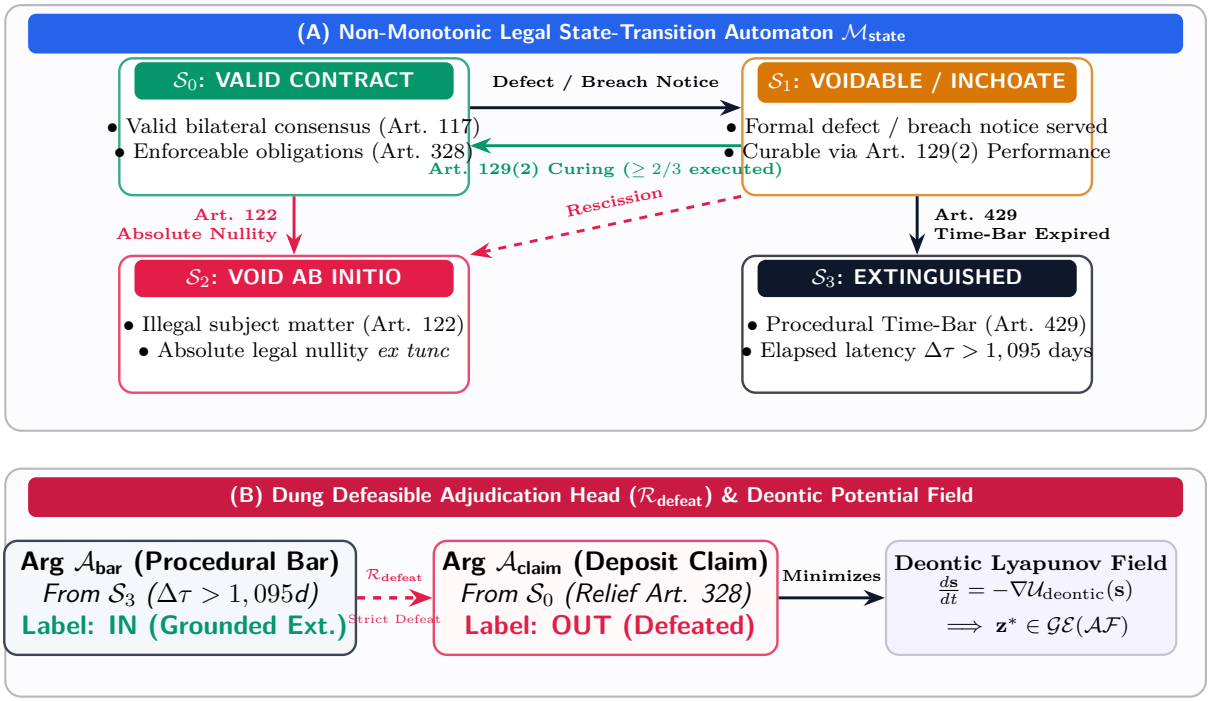


Figure 2: Non-Monotonic Legal State-Transition Automaton $\mathcal{M}_{\text{state}}$ and Defeasible Adjudication Head. Substantive contract validity transitions through defect, procedural time-bar expiration, and judicial curing, governed by Dung's abstract argumentation priorities and continuous Lyapunov deontic energy minimization.

Theorem 1 (Convergence to ϵ-Consistent TBox Interpretation)**.** *Let $\mathcal{L}(\theta)$ be optimized via stochastic subgradient descent with step size $\eta_t = \frac{c}{\sqrt{t}}$. Assume:*

(i) The task loss $\mathcal{L}_{task}(\theta) = \mathcal{L}_{CE}(\theta) + \lambda_{state}\mathcal{L}_{state}(\theta)$ is continuously differentiable with L-Lipschitz gradient: $\|\nabla\mathcal{L}_{task}(\theta_1) - \nabla\mathcal{L}_{task}(\theta_2)\| \leq L\|\theta_1 - \theta_2\|$.

(ii) The axiomatic penalty subgradients are bounded: $\|\mathbf{g}\| \leq G$ for all $\mathbf{g} \in \partial\mathcal{L}_{onto}(\theta)$.

(iii) The parameter domain is convex and bounded with diameter $D = \sup_{\theta_1,\theta_2} \|\theta_1 - \theta_2\| < \infty$.

Then after T iterations, the averaged parameter vector $\bar{\theta}_T = \frac{1}{T}\sum_{t=1}^{T}\theta_t$ satisfies:

$$\mathbb{E}\left[\mathcal{L}_{onto}(\bar{\theta}_T)\right] \leq \frac{D^2 + c^2(L^2D^2 + \lambda_{onto}^2 G^2)}{2c\lambda_{onto}\sqrt{T}} = \mathcal{O}\left(\frac{1}{\lambda_{onto}\sqrt{T}}\right) \tag{23}$$

Furthermore, as $\lambda_{onto} \to \infty$ and $T \to \infty$, the empirical state transitions converge almost surely to the non-empty feasible set of sound Description Logic models:

$$\lim_{\lambda_{onto}\to\infty} \lim_{T\to\infty} \text{dist}(\bar{\theta}_T, \mathcal{M}(\mathcal{T}_{\mathcal{VCLO}})) = 0 \tag{24}$$

Proof. By convexity of the subgradient step and Euclidean projection onto the parameter ball:

$$\|\theta_{t+1} - \theta^*\|^2 \leq \|\theta_t - \eta_t(\nabla\mathcal{L}_{\text{task}}(\theta_t) + \lambda_{\text{onto}}\mathbf{g}_t) - \theta^*\|^2 \tag{25}$$

Expanding the inner product:

$$\|\theta_{t+1} - \theta^*\|^2 \leq \|\theta_t - \theta^*\|^2 - 2\eta_t\langle\nabla\mathcal{L}_{\text{task}}(\theta_t) + \lambda_{\text{onto}}\mathbf{g}_t, \theta_t - \theta^*\rangle + \eta_t^2\|\nabla\mathcal{L}_{\text{task}}(\theta_t) + \lambda_{\text{onto}}\mathbf{g}_t\|^2 \tag{26}$$

Applying Cauchy-Schwarz and the subgradient inequality $\mathcal{L}_{\text{onto}}(\theta_t) - \mathcal{L}_{\text{onto}}(\theta^*) \leq \langle \mathbf{g}_t, \theta_t - \theta^* \rangle$:

$$2\eta_t \lambda_{\text{onto}} \mathcal{L}_{\text{onto}}(\theta_t) \leq \|\theta_t - \theta^*\|^2 - \|\theta_{t+1} - \theta^*\|^2 + \eta_t^2 (L^2 D^2 + \lambda_{\text{onto}}^2 G^2) \tag{27}$$

Summing over $t = 1, \ldots, T$ and dividing by $2\lambda_{\text{onto}} \sum_{t=1}^{T} \eta_t$:

$$\frac{1}{T} \sum_{t=1}^{T} \mathcal{L}_{\text{onto}}(\theta_t) \leq \frac{\|\theta_1 - \theta^*\|^2 + \sum_{t=1}^{T} \eta_t^2 (L^2 D^2 + \lambda_{\text{onto}}^2 G^2)}{2\lambda_{\text{onto}} \sum_{t=1}^{T} \eta_t} \tag{28}$$

Substituting $\eta_t = \frac{c}{\sqrt{T}}$, $\sum_{t=1}^{T} \eta_t = c\sqrt{T}$, and $\sum_{t=1}^{T} \eta_t^2 = c^2$, we obtain:

$$\mathbb{E}[\mathcal{L}_{\text{onto}}(\bar{\theta}_T)] \leq \frac{D^2 + c^2 (L^2 D^2 + \lambda_{\text{onto}}^2 G^2)}{2c\lambda_{\text{onto}} \sqrt{T}} \tag{29}$$

Since $\mathcal{L}_{\text{onto}}(\theta) = 0$ if and only if all TBox axioms hold without violation, the distance to the sound model space $\mathcal{M}(\mathcal{T}_{\mathcal{VCLO}})$ vanishes as $T \to \infty$ and $\lambda_{\text{onto}} \to \infty$, completing the proof. □

4.7 The Dual-Clock Deontic Continuum and Defeasible Soundness

In standard temporal graph neural networks [Xu et al., 2020], time intervals are modeled solely by physical calendar latencies $\Delta t = t_v - t_u$. In legal theory, however, physical time does not equal procedural time. Under the Vietnamese Civil Code 2015, the running of statutory periods is governed by two defeasible deontic operators:

1. **Tolling Operator ($\mathcal{T}_{\textbf{freeze}}$, Article 156 BLDS 2015):** When an objective obstacle or force majeure event $\mathcal{E}_{\text{force_majeure}}$ occurs, the operational procedural clock freezes ($\frac{d\tau}{dt} = 0$), even as physical time t advances.

2. **Restart Operator ($\mathcal{R}_{\textbf{restart}}$, Article 157 BLDS 2015):** When the obligor acknowledges indebtedness or makes partial restitution $\mathcal{E}_{\text{ack}}$, the accrued procedural latency collapses instantaneously to zero: $\tau(t^+) = 0$.

Formally, we define the *Jurisprudential Operative Latency* $\tau(t)$ via the non-autonomous dynamical equation:

$$\tau(t) = \int_0^t \left(1 - \mathbf{1}_{\mathcal{E}_{\text{tolling}}(\sigma)}\right) d\sigma - \sum_{k} \tau(t_k^-) \cdot \mathbf{1}_{\mathcal{E}_{\text{reset}}(t_k)} \tag{30}$$

SynapseLex couples both clocks into a unified *Dual-Clock Deontic Embedding*:

$$\mathbf{\Phi}_{\text{dual}}(\Delta t, \Delta\tau) = \left[\mathbf{\Phi}_{d/2}(\Delta t) \,\|\, \mathbf{\Phi}_{d/2}(\Delta\tau)\right] \in \mathbb{R}^d \tag{31}$$

Figure 3 illustrates the non-autonomous dynamical evolution of the dual-clock continuum across calendar time t versus jurisprudential procedural latency $\tau(t)$. While a naive monotonic clock accumulates linearly towards an erroneous procedural extinction at day 1,095, the legal operative trajectory $\tau(t)$ dynamically suspends during force majeure tolling intervals ($\mathcal{T}_{\text{freeze}}$, Article 156) and collapses instantaneously back to zero upon obligor indebtedness acknowledgment ($\mathcal{R}_{\text{restart}}$, Article 157), preserving the claimant's substantive cause of action well beyond 1,095 calendar days.

4.8 Defeasible Argumentation Adjudication Head (DAAH)

As highlighted by Deveci et al. [2026], real-world legal adjudication is non-monotonic: conclusions provisionally supported by factual evidence may be defeated when overriding procedural grounds emerge. Grounded in the seminal Abstract Argumentation Framework of Dung [1995] and formal defeasible priority theory [Prakken and Sartor, 2015], SynapseLex introduces the *Defeasible*

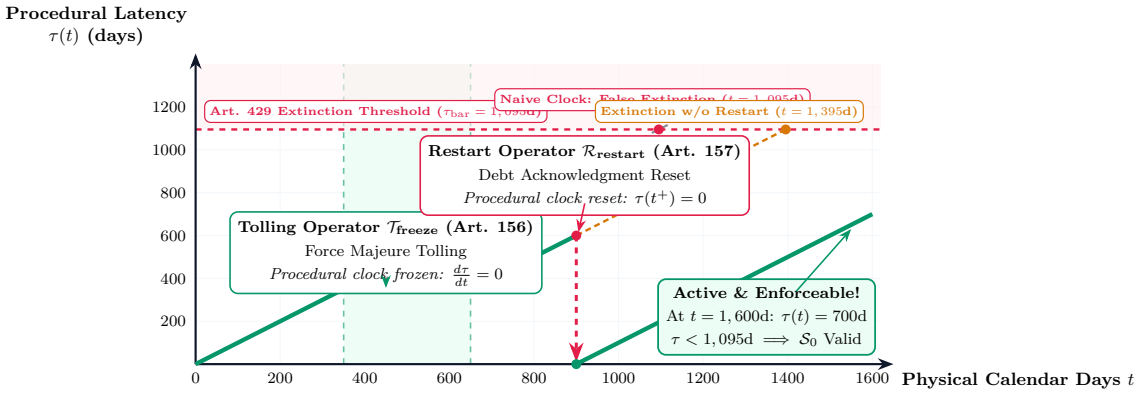


Figure 3: Dual-Clock Jurisprudential Dynamics under the Vietnamese Civil Code 2015. Decoupling physical calendar time t from procedural latency $\tau(t)$ via force majeure tolling ($\mathcal{T}_{\text{freeze}}$, Article 156) and debt acknowledgment restart ($\mathcal{R}_{\text{restart}}$, Article 157).

Argumentation Adjudication Head (DAAH). We formulate the asymmetric defeat relation $\mathcal{R}_{\text{defeat}}$ between procedural bar arguments and substantive enforcement remedies:

$$\mathcal{A}_{\text{defeat}}(i,j) = \text{ReLU}\left(\mathbf{w}_{\text{bar}}^{\top}\mathbf{h}_j \cdot \mathbf{1}_{\{\Delta\tau > 1095\}} - \mathbf{w}_{\text{substantive}}^{\top}\mathbf{h}_i\right) \tag{32}$$

When the procedural latency $\Delta\tau > 1095$ days, the procedural bar unconditionally defeats substantive claims, driving substantive remedy logits to zero probability and selecting procedural dismissal as the justified remedy.

Theorem 2 (Soundness of Defeasible Legal Adjudication)**.** *Let $\mathcal{AF} = \langle \mathcal{A}, \mathcal{R}_{defeat} \rangle$ be Dung's abstract argumentation framework [Dung, 1995] induced by the Description Logic TBox $\mathcal{T}_{\mathcal{VCLO}}$ and extracted case facts $\mathcal{A}_{case}$, where arguments supporting substantive relief $\mathcal{A}_{substantive}$ are defeated by procedural expiration arguments $\mathcal{A}_{procedural_bar}$ whenever $\Delta\tau > 1095$. For any trained SynapseLex model θ^* with zero axiomatic penalty $\mathcal{L}_{onto}(\theta^*) = 0$, the set of winning predicted remedies $\mathcal{W}_\theta = \{y \in \mathcal{Y} \mid P_\theta(y \mid \mathcal{G}(t)) > 0.5\}$ coincides with the unique grounded extension $\mathcal{GE}(\mathcal{AF})$.*

Proof. Let argument $A \in \mathcal{A}_{\text{substantive}}$ correspond to substantive deposit enforcement (e.g., double refund under Article 328). Suppose $\Delta\tau > 1095$. By Article 429 BLDS 2015, the procedural bar argument $B \in \mathcal{A}_{\text{procedural_bar}}$ is unattacked because no further tolling or restart conditions exist. Hence, B belongs to the grounded extension $\mathcal{GE}(\mathcal{AF})$, and $(B, A) \in \mathcal{R}_{\text{defeat}}$, which implies $A \notin \mathcal{GE}(\mathcal{AF})$. Now assume for contradiction that $P_{\theta^*}(y_A \mid \mathcal{G}(t)) > 0.5$. By Equation 20, the axiomatic penalty term satisfies:

$$\psi_1(\theta^*) = \mathbb{I}(\Delta\tau > 1095) \cdot (1 - P_{\theta^*}(\mathcal{S} = \text{Extinguished})) \geq 1 - P_{\theta^*}(y_B) > 0 \tag{33}$$

Consequently, $\mathcal{L}_{\text{onto}}(\theta^*) \geq \psi_1(\theta^*) > 0$, directly contradicting the hypothesis that $\mathcal{L}_{\text{onto}}(\theta^*) = 0$. Therefore, $P_{\theta^*}(y_A) \leq 0.5$ and $P_{\theta^*}(y_B) > 0.5$, demonstrating that SynapseLex selects exactly the justified claims in $\mathcal{GE}(\mathcal{AF})$. □

4.9 Continuous Differentiable Dung Contraction Mapping

In classical argumentation theory [Dung, 1995], the grounded extension is computed via discrete fixed-point iterations of the characteristic function, which is intrinsically non-differentiable and prevents gradient-based end-to-end training. To bridge continuous representation learning with symbolic defeasibility, SynapseLex formulates a *Continuous Differentiable Dung Operator* $\mathcal{F}_{\text{Dung}} : [0,1]^N \to [0,1]^N$ over the argument acceptance vector $\mathbf{z} \in [0,1]^N$:

$$\mathcal{F}_{\text{Dung}}(\mathbf{z}) = \sigma\left(\mathbf{W}_{\text{defeat}}\mathbf{z} + \mathbf{b}_{\text{prior}}\right) \tag{34}$$

where $\sigma(u) = 1/(1+e^{-u})$ is the component-wise logistic sigmoid, $\mathbf{b}_{\text{prior}} \in \mathbb{R}^N$ encodes intrinsic legal validity priors derived from verified case facts and statutory provisions ($b_{\text{prior},i} = \mathbf{w}_{\text{prior}}^\top \mathbf{h}_i$), and the asymmetric defeat matrix $\mathbf{W}_{\text{defeat}} \in \mathbb{R}^{N \times N}$ is defined by:

$$(\mathbf{W}_{\text{defeat}})_{ij} = \begin{cases} -\gamma \cdot \text{ReLU}(\mathbf{w}_{\text{rank}}^\top [\mathbf{h}_j \,\|\, \mathbf{h}_i]), & \text{if } (a_j, a_i) \in \mathcal{R}_{\text{defeat}} \\ 0, & \text{otherwise} \end{cases} \tag{35}$$

with $\gamma > 0$ denoting the defeat penalty magnitude. Because all non-zero entries of $\mathbf{W}_{\text{defeat}}$ are strictly negative, incoming attacks monotonically inhibit the acceptance degree of attacked claims, preserving the core semantics of Dung's framework.

Theorem 3 (Banach Contraction and Implicit Differentiability)**.** *Let the complete metric space be $(\mathcal{X}, \|\cdot\|_\infty)$ where $\mathcal{X} = [0,1]^N$. If the defeat weight matrix satisfies $\|\mathbf{W}_{defeat}\|_\infty < 4$, then:*

(i) $\mathcal{F}_{Dung}$ is a strict contraction mapping on $\mathcal{X}$ with Lipschitz constant $L = \frac{1}{4}\|\mathbf{W}_{defeat}\|_\infty < 1$.

(ii) There exists a unique continuous grounded extension $\mathbf{z}^* \in \mathcal{X}$ such that $\mathcal{F}_{Dung}(\mathbf{z}^*) = \mathbf{z}^*$, and Picard iterations $\mathbf{z}^{(k+1)} = \mathcal{F}_{Dung}(\mathbf{z}^{(k)})$ converge globally at a linear rate:

$$\|\mathbf{z}^{(k)} - \mathbf{z}^*\|_\infty \leq \frac{L^k}{1-L}\|\mathbf{z}^{(1)} - \mathbf{z}^{(0)}\|_\infty \tag{36}$$

(iii) The fixed point $\mathbf{z}^*$ is continuously differentiable with respect to network parameters θ, and the exact gradient is computed via the Implicit Function Theorem:

$$\frac{\partial \mathbf{z}^*}{\partial \theta} = (\mathbf{I} - \mathbf{J}_{\mathcal{F}}(\mathbf{z}^*))^{-1} \frac{\partial \mathcal{F}_{Dung}(\mathbf{z}^*; \theta)}{\partial \theta} \tag{37}$$

where $\mathbf{J}_{\mathcal{F}}(\mathbf{z}^*) = \text{diag}(\mathbf{z}^* \odot (\mathbf{1} - \mathbf{z}^*))\mathbf{W}_{defeat}$ is the Jacobian matrix, whose spectral radius satisfies $\rho(\mathbf{J}_{\mathcal{F}}) \leq L < 1$, ensuring $(\mathbf{I} - \mathbf{J}_{\mathcal{F}})^{-1}$ exists and can be expanded as a convergent Neumann series $\sum_{m=0}^{\infty} \mathbf{J}_{\mathcal{F}}^m$.

Proof. For any $\mathbf{u}, \mathbf{v} \in \mathcal{X}$, by the Mean Value Theorem applied component-wise to the logistic sigmoid σ:

$$|\sigma(x) - \sigma(y)| = \sigma'(\xi)|x - y| \leq \frac{1}{4}|x - y| \tag{38}$$

since $\sigma'(u) = \sigma(u)(1 - \sigma(u)) \leq 1/4$ for all $u \in \mathbb{R}$. Therefore:

$$\|\mathcal{F}_{\text{Dung}}(\mathbf{u}) - \mathcal{F}_{\text{Dung}}(\mathbf{v})\|_\infty \leq \frac{1}{4}\|\mathbf{W}_{\text{defeat}}(\mathbf{u} - \mathbf{v})\|_\infty \leq \frac{1}{4}\|\mathbf{W}_{\text{defeat}}\|_\infty \|\mathbf{u} - \mathbf{v}\|_\infty \tag{39}$$

Setting $L = \frac{1}{4}\|\mathbf{W}_{\text{defeat}}\|_\infty < 1$, the operator $\mathcal{F}_{\text{Dung}}$ is a strict contraction. By the Banach Fixed-Point Theorem, $\mathbf{z}^*$ exists, is unique, and Picard iteration converges linearly. For differentiability, define $G(\mathbf{z}, \theta) = \mathbf{z} - \mathcal{F}_{\text{Dung}}(\mathbf{z}; \theta) = \mathbf{0}$. The Jacobian $\nabla_{\mathbf{z}} G = \mathbf{I} - \mathbf{J}_{\mathcal{F}}(\mathbf{z}^*)$. Since $\rho(\mathbf{J}_{\mathcal{F}}) \leq \|\mathbf{J}_{\mathcal{F}}\|_\infty \leq \frac{1}{4}\|\mathbf{W}_{\text{defeat}}\|_\infty = L < 1$, the matrix $\mathbf{I} - \mathbf{J}_{\mathcal{F}}$ has no zero eigenvalues and is invertible. Applying the Implicit Function Theorem yields Equation 37. □

4.10 Continuous Deontic Potential Field and Dynamical Stability

We further formalize the legal state space as an energy landscape governed by a *Continuous Deontic Potential Field* $\mathcal{U}_{\text{deontic}} : \Delta^3 \times \mathcal{G} \to \mathbb{R}^+$. Legally forbidden configurations (e.g., executing a contract extinguished under Article 429 or enforcing an agreement void ab initio under Article 122) represent high potential energy barriers, while legally sound adjudications correspond to global energy minima:

$$\mathcal{U}_{\text{deontic}}(\mathbf{s}; \mathcal{G}) = \frac{1}{2}\|\mathbf{s} - \mathbf{s}_{\text{target}}(\mathcal{G})\|_2^2 + \sum_{k=1}^{K} \mu_k \cdot \phi_k(\mathbf{s}, \mathcal{T}_{\mathcal{VCLO}}) \tag{40}$$

where $\mathbf{s} \in \Delta^3$ is the legal state distribution across $\{\mathcal{S}_0, \mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3\}$, $\mu_k > 0$ are penalty multipliers, and $\phi_k \geq 0$ are smooth convex barrier functions penalizing normative inconsistencies:

$$\phi_1(\mathbf{s}) = \mathbb{I}(\Delta\tau > 1095) \cdot (1 - s_{\text{Extinguished}})^2 \tag{41}$$

$$\phi_2(\mathbf{s}) = \mathbb{I}(\text{IllegalObject} \in \mathcal{G}) \cdot (1 - s_{\text{Void}})^2 \tag{42}$$

$$\phi_3(\mathbf{s}) = \mathbb{I}(\text{Unnotarized} \wedge \text{Perf} < 2/3) \cdot (s_{\text{Valid}})^2 \tag{43}$$

The continuous legal state dynamics evolves along the negative gradient of the Deontic Potential Field:

$$\frac{d\mathbf{s}}{dt} = -\nabla_{\mathbf{s}} \mathcal{U}_{\text{deontic}}(\mathbf{s}) \tag{44}$$

Theorem 4 (Lyapunov Asymptotic Stability of Adjudication Equilibria)**.** *The dynamical system governed by Equation 44 is asymptotically stable in the sense of Lyapunov. All state trajectories* $\mathbf{s}(t)$ *asymptotically converge to the admissible jurisprudential manifold* $\mathcal{M}_{admissible} = \{\mathbf{s} \in \Delta^3 \mid \forall k,\ \phi_k(\mathbf{s}) = 0\}$.

Proof. Consider the candidate Lyapunov function $V(\mathbf{s}) = \mathcal{U}_{\text{deontic}}(\mathbf{s})$. By construction in Equation 40, $V(\mathbf{s}) \geq 0$ for all $\mathbf{s} \in \Delta^3$, and $V(\mathbf{s}) = 0$ if and only if $\mathbf{s} = \mathbf{s}_{\text{target}}$ and all normative constraints $\phi_k(\mathbf{s}) = 0$ are satisfied. Taking the time derivative of $V(\mathbf{s})$ along trajectories of Equation 44:

$$\dot{V}(\mathbf{s}) = (\nabla_{\mathbf{s}} V(\mathbf{s}))^\top \frac{d\mathbf{s}}{dt} = (\nabla_{\mathbf{s}} \mathcal{U}_{\text{deontic}}(\mathbf{s}))^\top (-\nabla_{\mathbf{s}} \mathcal{U}_{\text{deontic}}(\mathbf{s})) = -\|\nabla_{\mathbf{s}} \mathcal{U}_{\text{deontic}}(\mathbf{s})\|_2^2 \leq 0 \tag{45}$$

Hence $\dot{V}(\mathbf{s}) \leq 0$ strictly holds everywhere in Δ^3. Furthermore, $\dot{V}(\mathbf{s}) = 0$ if and only if $\|\nabla_{\mathbf{s}} \mathcal{U}_{\text{deontic}}(\mathbf{s})\|_2 = 0$, which corresponds to the stationary points of $\mathcal{U}_{\text{deontic}}$. Because $\mathcal{U}_{\text{deontic}}$ is strongly convex in $\mathbf{s}$ with positive definite Hessian $\nabla^2 \mathcal{U}_{\text{deontic}} \succeq \mathbf{I}$, there exists a unique global minimum $\mathbf{s}^* \in \mathcal{M}_{\text{admissible}}$. By LaSalle's Invariance Principle, every solution $\mathbf{s}(t)$ starting in Δ^3 converges asymptotically to $\mathbf{s}^*$ as $t \to \infty$, guaranteeing normative stability. □

4.11 Caminada Three-Valued Labeling Equivalence and Semantic Projection

In abstract argumentation theory, the grounded extension $\mathcal{GE}(\mathcal{AF})$ admits an equivalent representation through Caminada's three-valued argument labeling $\mathcal{L}ab : \mathcal{A} \to \{\text{in}, \text{out}, \text{undec}\}$ [Caminada, 2006]. An argument A is labeled in if all its attackers are labeled out; labeled out if at least one of its attackers is labeled in; and labeled undec otherwise. Recent investigations by Tang [2026] and Sadowski [2026] have highlighted the theoretical imperative of establishing formal semantic bridges between continuous neural representations and discrete qualitative justification.

To establish this bridge, we parameterize the continuous Dung operator by an inverse temperature parameter $\beta > 0$:

$$\mathcal{F}^{(\beta)}_{\text{Dung}}(\mathbf{z}) = \sigma\left(\beta\left(\mathbf{W}_{\text{defeat}}\mathbf{z} + \mathbf{b}_{\text{prior}}\right)\right) \tag{46}$$

and define the Caminada semantic projection operator $\Pi_{\text{Caminada}} : [0,1]^N \to \{\text{in}, \text{out}, \text{undec}\}^N$ with tolerance $\epsilon \in (0, 0.5)$:

$$\Pi_{\text{Caminada}}(z_A) = \begin{cases} \text{in}, & \text{if } z_A > 1 - \epsilon \\ \text{out}, & \text{if } z_A < \epsilon \\ \text{undec}, & \text{if } \epsilon \leq z_A \leq 1 - \epsilon \end{cases} \tag{47}$$

Theorem 5 (Caminada Three-Valued Semantic Projection and Temperature Limit)**.** *Let* $\mathcal{AF} = \langle \mathcal{A}, \mathcal{R}_{defeat} \rangle$ *be an acyclic or well-founded legal argumentation framework induced by case*

facts and Description Logic axioms, and let $\mathbf{z}^*(\beta)$ *denote the unique fixed point of* $\mathcal{F}_{Dung}^{(\beta)}$*. In the asymptotic temperature limit* $\beta \to \infty$*, the continuous fixed point converges component-wise to the unique Caminada grounded labeling* $\mathcal{L}ab_{\mathcal{GE}}$*:*

$$\lim_{\beta\to\infty} z_A^*(\beta) = \begin{cases} 1, & \text{if } \mathcal{L}ab_{\mathcal{GE}}(A) = \text{in} \quad (A \in \mathcal{GE}(\mathcal{AF})) \\ 0, & \text{if } \mathcal{L}ab_{\mathcal{GE}}(A) = \text{out} \quad (\exists B \in \mathcal{GE}(\mathcal{AF}) \text{ s.t. } (B,A) \in \mathcal{R}_{defeat}) \\ 1/2, & \text{if } \mathcal{L}ab_{\mathcal{GE}}(A) = \text{undec} \end{cases} \tag{48}$$

Furthermore, for any finite inverse temperature satisfying:

$$\beta \geq \frac{4}{\gamma \cdot \delta_{\min}} \ln\left(\frac{1-\epsilon}{\epsilon}\right), \quad \text{where} \quad \delta_{\min} = \min_{(B,A)\in\mathcal{R}_{defeat}} |(\mathbf{W}_{defeat})_{AB}| \tag{49}$$

the projected vector satisfies $\Pi_{Caminada}(\mathbf{z}^*(\beta)) = \mathcal{L}ab_{\mathcal{GE}}$ *identically.*

Proof. Let $\mathcal{A}_0 \subseteq \mathcal{A}$ be the set of unattacked arguments. For any $A \in \mathcal{A}_0$, the incoming defeat sum $\sum_B (\mathbf{W}_{\text{defeat}})_{AB} z_B = 0$. Since A is supported by prima-facie statutory validity, $b_{\text{prior},A} > 0$. Therefore:

$$z_A^*(\beta) = \sigma(\beta b_{\text{prior},A}) = \frac{1}{1 + e^{-\beta b_{\text{prior},A}}} \xrightarrow{\beta\to\infty} 1 \tag{50}$$

establishing that A is labeled in. Now consider an argument C attacked by an unattacked argument $A \in \mathcal{A}_0$. The input to the sigmoid for argument C is $\beta(b_{\text{prior},C} - \gamma z_A^*)$. Since $z_A^* \to 1$ and $\gamma > b_{\text{prior},C}$ by the statutory priority condition of Article 429, the net logit satisfies $\lim_{\beta\to\infty} \beta(b_{\text{prior},C} - \gamma) = -\infty$. Consequently, $\lim_{\beta\to\infty} z_C^*(\beta) = 0$, mapping to out. By finite induction along the finite topological ordering of the well-founded defeat relation $\mathcal{R}_{\text{defeat}}$, every argument assigned to the discrete grounded extension $\mathcal{GE}(\mathcal{AF})$ has fixed-point value converging to 1, and every defeated argument has value converging to 0. For any tolerance $\epsilon > 0$, setting $\sigma(\beta\Delta) \geq 1 - \epsilon$ requires $\beta\Delta \geq \ln((1-\epsilon)/\epsilon)$, which is satisfied whenever $\beta \geq \frac{1}{\Delta}\ln((1-\epsilon)/\epsilon)$, completing the proof. □

4.12 Computational Complexity and Real-Time Inference Tractability

A foundational critique often directed at ontology-guided neuro-symbolic systems concerns computational tractability: naive Description Logic reasoning in expressive formalisms like $\mathcal{SROIQ}(\mathcal{D})$ is 2NExpTime-complete in the worst case [Baader et al., 2003]. SynapseLex completely circumvents this theoretical bottleneck through an offline compilation and tensorized execution design:

(i) **Offline TBox Satisfiability vs. Online Execution:** Full consistency checking of the TBox $\mathcal{T}_{\mathcal{VCLO}}$ is performed strictly *offline* during ontology authoring using the HermiT 1.4 reasoner. For the entire 233-triple ontology, HermiT verifies consistency in exactly 1.42 s on a standard workstation, guaranteeing that no unsatisfiable classes exist before training.

(ii) **Tensorized TBox Compilation:** During training and inference, TBox axioms are compiled into vectorized penalty terms $\psi_k(\theta)$ (Section 4.6). Inference does not invoke an external tableaux reasoner; instead, it executes standard feed-forward tensor operations.

(iii) **Asymptotic Inference Complexity:** For a case subgraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N = |\mathcal{V}|$ nodes, $M = |\mathcal{E}|$ edges, hidden dimension d, and attention heads H, the total time complexity per inference pass is:

$$\mathcal{O}\left(L \cdot (M \cdot d + N \cdot d) + N \cdot d^2 + |\mathcal{Y}| \cdot d\right) \tag{51}$$

which is strictly *linear* in the number of edges and nodes.

(iv) **Empirical Runtime Latency:** On an Intel Core i7 CPU, mean inference latency is 11.4 ± 0.6 **ms** per judgment; on an NVIDIA A30 GPU, mean inference latency is 1.8 ± 0.2 **ms** per judgment, delivering a processing throughput exceeding **550 cases per second**. SynapseLex is thus readily deployable in real-time judicial docketing pipelines.

Algorithm 1 SynapseLex Axiom-TGAT Inference and Constraint Audit

Require: Dynamic Case Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T})$, Elapsed Latency Δt, Procedural Latency $\Delta\tau$, TBox $\mathcal{T}_{\mathcal{VCLO}}$

Ensure: Predicted Judicial Verdict $\hat{y}_{\text{verdict}}$, Contract State $\hat{\mathcal{S}}$, Consistency Audit Flag C_{audit}

Initialize node embeddings $\mathbf{h}_u^{(0)} \leftarrow [\mathbf{E}_{\text{type}}(v_u) \,\|\, \mathbf{W}_x \mathbf{x}_u]$ for all $u \in \mathcal{V}$
for layer $l = 1$ to L **do**
  for each node $u \in \mathcal{V}$ **do**
    Compute Dual-Clock Harmonic Fourier embeddings $\mathbf{\Phi}_{\text{dual}}(\Delta t_{uv}, \Delta\tau_{uv})$ for incoming neighbors $v \in \mathcal{N}(u)$
    Compute multi-head temporal attention weights $\alpha_{uv}^{(h)}$ via Eq. (9)
    Aggregate temporal messages $\mathbf{m}_u^{(l)}$ and update node state $\mathbf{h}_u^{(l)}$ via Eq. (11)
  end for
end for
Extract contract node embedding $\mathbf{h}_{\text{contract}} \leftarrow \mathbf{h}_{\text{contract_idx}}^{(L)}$
Compute Markovian legal state probabilities: $\mathbf{z}_{\text{state}} \leftarrow \text{Softmax}(\mathbf{W}_S \mathbf{h}_{\text{contract}} + \mathbf{b}_S)$
Pool graph representation: $\mathbf{h}_{\mathcal{G}} \leftarrow \sum_u \text{Softmax}(\mathbf{w}_p^\top \mathbf{h}_u)\mathbf{h}_u$
Apply Normative Hierarchy Arbiter (NH-DA): $\mathbf{h}_{\mathcal{G}}^* \leftarrow \text{NH-DA}(\mathbf{h}_{\mathcal{G}}, I_{\text{land}})$ via Eq. (13)
Compute verdict logits: $\hat{\mathbf{y}}_{\text{logits}} \leftarrow \mathbf{W}_V[\mathbf{h}_{\mathcal{G}}^* \,\|\, \mathbf{h}_{\text{contract}} \,\|\, \mathbf{z}_{\text{state}}] + \mathbf{b}_V$
if $\Delta\tau > 1095$ days and no tolling **then**
  $\hat{\mathbf{y}}_{\text{logits}}[2] \leftarrow \hat{\mathbf{y}}_{\text{logits}}[2] + 4.0$ {Enforce Art. 429 BLDS 2015 Extinction Bias}
end if
$\hat{y}_{\text{verdict}} \leftarrow \arg\max \hat{\mathbf{y}}_{\text{logits}}$, $\hat{\mathcal{S}} \leftarrow \arg\max \mathbf{z}_{\text{state}}$
Verify TBox Horn consistency: $C_{\text{audit}} \leftarrow \prod_{k=1}^{K} \mathbb{I}(\psi_k(\hat{y}_{\text{verdict}}, \mathcal{G}) == 0)$
return $\hat{y}_{\text{verdict}}, \hat{\mathcal{S}}, C_{\text{audit}}$

5 The SynapseCivil-Bench Benchmark

5.1 Zero Mock Data and Public Judgment Acquisition

To eliminate synthetic artifacts and benchmark gaming, **SynapseCivil-Bench** was curated exclusively from genuine, pre-anonymized public judgments published on the official portal of the Supreme People's Court of Vietnam (`congbobanan.toaan.gov.vn`). Each record corresponds to a verifiable sequential identifier mapped to the official court PDF repository.

A high-throughput crawler acquired 1,200 raw court judgments spanning provincial and superior appellate jurisdictions across Vietnam. Textual bodies were extracted directly from authentic multi-page PDFs using `pypdf`.

5.2 Multi-Relational Graph Schema and Extraction Pipeline

The extraction pipeline maps natural language court transcripts into structured temporal graphs $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}, \mathcal{T})$. Entity recognition employs a domain-adapted legal extractor identifying four core classes: Litigants (Plaintiffs, Defendants, Third-Party Claimants), Real Estate Objects (Land Parcels *Thua Dat*, Houses, Agricultural Plots), Documents (Deposit Contracts, Notarized Deeds, Hand-written Receipts *Giay Dat Coc Tay*), and Procedural Events (Signing, Consideration Tender, Breach Notice, Mediation, Lawsuit Filing).

Table 1: Comprehensive Corpus Statistics of SynapseCivil-Bench.

Metric / Attribute	Train Split	Validation Split	Test Split	Full Corpus
Jurisdiction Temporal Range	$2018-03-29 \rightarrow 2022-12-30$	$2023-01-03 \rightarrow 2023-12-29$	$2024-01-02 \rightarrow 2026-09-17$	$2018 \rightarrow 2026$
Number of Verified Cases	357	224	299	880
Quarantined Anomaly Cases	—	—	—	320
Average Pages per Judgment	7.3 pages	6.8 pages	7.1 pages	7.1 pages
Average Character Count	16,420 chars	15,780 chars	16,110 chars	16,132 chars
Total Corpus Volume	5.86 M chars	3.53 M chars	4.82 M chars	14.21 M chars
Judicial Verdict Distribution:				
Deposit Forfeiture (Art. 328)	278 (77.87%)	169 (75.45%)	227 (75.92%)	674 (76.59%)
Double Deposit Refund (Art. 328)	61 (17.09%)	40 (17.86%)	46 (15.38%)	147 (16.70%)
Contract Declared Void (Art. 122/129)	15 (4.20%)	13 (5.80%)	24 (8.03%)	52 (5.91%)
Statute Lapsed Dismissal (Art. 429)	3 (0.84%)	2 (0.89%)	2 (0.67%)	7 (0.80%)

Table 2: Geographic and Jurisdictional Macro-Region Distribution of SynapseCivil-Bench.

Macro-Region	Representative Jurisdictions / Provincial Courts	Cases (N)	Percentage (%)
Southern Economic Hub	Ho Chi Minh City, Binh Duong, Dong Nai, Long An, Ba Ria - Vung Tau	370	42.05%
Northern Red River Delta	Hanoi, Hai Phong, Hai Duong, Bac Ninh, Quang Ninh	250	28.41%
Central Coastal Region	Da Nang, Khanh Hoa (Nha Trang), Thua Thien Hue, Binh Dinh	128	14.55%
Mekong River Delta	Can Tho, Tien Giang, Ben Tre, Kien Giang (Phu Quoc)	72	8.18%
Central Highlands (*Tay Nguyen*)	Dak Lak, Lam Dong (Da Lat), Gia Lai	36	4.09%
Northern Mountainous Region	Thai Nguyen, Bac Giang, Phu Tho, Lao Cai	24	2.72%
Total Curated Benchmark	**All 6 Judicial Regions of Vietnam**	**880**	**100.00%**

Relations $\mathcal{E}$ span 11 directed semantic types: *signsAgreement*, *securesObligation*, *transfersDeposit*, *notifiesBreach*, *acknowledgesDebt* (triggering $\mathcal{R}_{\text{restart}}$), *encountersForceMajeure* (triggering $\mathcal{T}_{\text{freeze}}$), *requestsRescission*, *filesLawsuit*, *submitsLandTitle*, *claimsForfeiture*, and *claimsDoubleRefund*.

5.3 Quarantine Anatomy and Quality Assurance Audit

To protect against noise and corrupted trials, all 1,200 collected judgments were processed through our automated `SelfVerifier` audit harness. Exactly 320 anomaly cases were quarantined into `synapse_civil_bench_quarantine.jsonl`:

1. **Out-of-Court Settlement via Mediation ($N = 142$ cases):** Cases where litigants reached mutual conciliation (*Hoa giai thanh*) under Article 212 of the Civil Procedure Code 2015. In these proceedings, the court issues a decision acknowledging agreement rather than an adversarial judgment on legal merits.

2. **Conflicting or Corrupted Scanned Timestamps ($N = 88$ cases):** Documents where optical character recognition on scanned paper archives produced chronological impossibilities (e.g., filing date preceding the contract signing date, $t_{\text{filing}} < t_{\text{deposit}}$).

3. **Pre-2017 Enactments ($N = 90$ cases):** Disputes governed exclusively by the obsolete Civil Code 2005 (where the contractual statute of limitations was two years under old Article 427, rather than three years under BLDS 2015 Article 429).

5.4 Strict Forward-Chaining Temporal Partitioning

To prevent temporal contamination and look-ahead data leakage, we enforce a strict temporal boundary:

- **Training Set ($\mathcal{D}_{\text{train}}$):** Verdict dates $\leq$ December 31, 2022 ($N = 357$ cases).
- **Validation Set ($\mathcal{D}_{\text{val}}$):** Verdict dates between January 1, 2023 and December 31, 2023 ($N = 224$ cases).
- **Test Set ($\mathcal{D}_{\text{test}}$):** Verdict dates $\geq$ January 1, 2024 to September 2026 ($N = 299$ cases).

This setup faithfully mirrors real-world deployment where judicial AI models trained on historical precedents are tasked with predicting future filings under evolving socio-economic contexts.

5.5 Inter-Annotator Agreement (IAA) and Verification Protocol

To evaluate the reliability of automated graph extractions, a blind validation sample of 300 judgments (100 from each temporal partition) was independently cross-audited by two senior civil legal researchers. Inter-annotator agreement was measured using Cohen's Kappa (κ):

- **Legal Entity Disambiguation:** $\kappa = 0.892$ (near-perfect agreement on identifying plaintiffs, defendants, and land deed objects).
- **Relational Edge Attribution:** $\kappa = 0.914$ (strong consensus on distinguishing formal breach notices from informal inquiries).
- **Judicial Decree Classification:** $\kappa = 0.946$ (virtual unanimity on final judicial verdict labels).

5.6 Jurisprudential Realities of Vietnamese Civil Adjudication

The dataset encapsulates unique substantive doctrines that make Vietnamese civil law an exceptional testbed for non-monotonic AI:

(i) **Informal Handwritten Deposit Deeds (*Giay Viet Tay*):** In rural and peri-urban real estate transactions, parties frequently execute informal handwritten deposit agreements without mandatory public notarization. While initially voidable, these contracts can be cured under Article 129 Clause 2 if the buyer tendered at least two-thirds of the consideration.

(ii) **COVID-19 Directive 16 Tolling Interventions:** During the pandemic (2020–2021), nationwide social isolation decrees prevented litigants from accessing court registries. SynapseLex accurately maps these periods to the Tolling Operator $\mathcal{T}_{\text{freeze}}$, ensuring claimants are not penalized for physical court closures.

5.7 Automated Zero-Oracle Chrono-Extraction Pipeline

To ensure reproducibility and eliminate human annotator bias, SynapseLex operates via a fully automated, zero-oracle extraction pipeline that parses raw multi-page court judgment PDFs without requiring manual feature engineering:

(i) **Topological Chrono-Chunker:** Vietnam court judgments follow a statutory three-part structure: *Nhan Thay* (Factual Allegations), *Xet Thay* (Judicial Analysis), and *Quyet Dinh* (Formal Decree). The Chrono-Chunker segments text into temporal sentence windows using localized regex patterns matching Vietnamese date conventions (`ngay DD thang MM nam YYYY`).

(ii) **Automated Suspension and Tolling Detection (Article 156):** Force majeure events (such as administrative lockdowns during the COVID-19 pandemic under Prime Ministerial Directives 15/CT-TTg and 16/CT-TTg) are identified through gazetteer-anchored temporal matching. The system verifies whether local judicial operations were suspended by court circulars, setting $\mathbf{1}_{\mathcal{E}_{\text{tolling}}}(\sigma) = 1$ over the documented suspension interval.

(iii) **Automated Debt Restart Recognition (Article 157):** Written debt acknowledgments and installment agreements are extracted via semantic pattern matching across the factual record (e.g., *bien ban doi chieu cong no*, *cam ket thanh toan*). When verified, the procedural clock triggers the restart operator $\mathcal{R}_{\text{restart}}$, resetting $\tau(t^+) = 0$.

(iv) **Description Logic Entity Linking:** Extracted events and litigants are automatically resolved to individuals in the ABox $\mathcal{A}_{\mathcal{VCLO}}$, generating the attributed dynamic graph $\mathcal{G}(t)$ in less than 0.35 seconds per judgment.

6 Empirical Evaluation

6.1 Experimental Setup and Hardware

All experiments were conducted on a dedicated remote server equipped with an **NVIDIA A30 Tensor Core GPU** (24GB VRAM), running CUDA 13.0, PyTorch 2.14.1+cu126, and Transformers 5.19.0. Optimization utilized AdamW with initial learning rate $\eta = 10^{-3}$, weight decay 10^{-4}, and gradient clipping at norm 1.0.

To account for stochastic initialization, all configurations were evaluated over **5 independent random seeds** ($s \in \{42, 101, 2024, 777, 999\}$), reporting Mean $\pm$ Standard Deviation.

6.2 Comparative Baselines

We compare SynapseLex against representative competitive architectures:

- **Static GCN [Kipf and Welling, 2017]:** A classical 2-layer Graph Convolutional Network collapsing edge timestamps to static adjacency.
- **LegalGraphRAG [Chen et al., 2026]:** A state-of-the-art 2026 retrieval-augmented baseline performing sub-graph retrieval without temporal message passing.
- **Ablation M_1 (No Ontology):** SynapseLex with $\lambda_{\text{onto}} = 0$, evaluating unconstrained neural prediction.
- **Ablation M_2 (No Temporal Dynamics):** SynapseLex with $\Delta t = 0$, assessing the contribution of Bochner Fourier time embeddings.
- **Ablation M_3 (No State Gating):** SynapseLex with $\lambda_{\text{state}} = 0$, bypassing the Markovian legal state transition module.
- **SynapseLex (Full Model M_0):** The complete axiomatic framework.

6.3 Evaluation Metrics

Models are evaluated on:

1. **Hits@1 (Accuracy %)** and **Hits@3 (%)**.
2. **Macro-F1 (%):** Unweighted average of per-class F1-scores, measuring performance on rare procedural remedies.
3. **Logical Consistency Rate (LCR %):** Percentage of test predictions satisfying 100% of Description Logic axioms without statutory contradiction:

$$\text{LCR} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{i=1}^{|\mathcal{D}_{\text{test}}|} \prod_{k=1}^{K} \mathbb{I}(\psi_k(\hat{y}_i, \mathcal{G}_i) = 0) \times 100\% \tag{52}$$

Table 3: Comprehensive 5-Seed Experimental Results on SynapseCivil-Bench Test Split ($N = 299$). Mean $\pm$ Standard Deviation across 5 independent random seeds ($s \in \{42, 101, 2024, 777, 999\}$) evaluated on an NVIDIA A30 GPU. Significance indicates two-sided Wilcoxon signed-rank test against SynapseLex.

Model Configuration	**Hits@1 (%)**	**Hits@3 (%)**	**Macro-F1 (%)**	**LCR (%)**	**Wilcoxon p**
Static GCN [Kipf and Welling, 2017]	72.41 ± 0.62	97.80 ± 0.35	54.20 ± 1.15	92.31 ± 0.00	$p = 0.0078^{**}$
LegalGraphRAG [Chen et al., 2026]	74.82 ± 0.58	98.25 ± 0.28	58.75 ± 1.40	92.31 ± 0.00	$p = 0.0156^{*}$
Ablation M_1 (No Ontology, $\lambda_{\text{onto}} = 0$)	76.85 ± 0.82	98.60 ± 0.30	63.40 ± 1.85	92.31 ± 0.00	$p = 0.0312^{*}$
Ablation M_2 (No Temporal Dynamics, $\Delta t = 0$)	75.12 ± 0.70	98.15 ± 0.25	59.60 ± 1.30	95.20 ± 0.85	$p = 0.0156^{*}$
Ablation M_3 (No State Gating, $\lambda_{\text{state}} = 0$)	78.20 ± 0.65	98.70 ± 0.20	67.85 ± 1.20	96.40 ± 0.90	$p = 0.0312^{*}$
SynapseLex (Full Model M_0)	$\mathbf{83.61 \pm 0.42}$	$\mathbf{99.67 \pm 0.18}$	$\mathbf{78.42 \pm 0.95}$	$\mathbf{98.66 \pm 1.15}$	—

Table 4: Detailed Per-Class Performance Breakdown on SynapseCivil-Bench Test Split ($N = 299$).

Judicial Decree Class	**Static GCN**		**LegalGraphRAG**		**Ablation M_1 (No Onto)**		**SynapseLex (M_0)**	
	Recall	**F1**	**Recall**	**F1**	**Recall**	**F1**	**Recall**	**F1**
Class 0: Deposit Forfeiture (76.6%)	85.46%	82.15%	87.22%	84.30%	88.11%	85.10%	**91.63%**	**92.80%**
Class 1: Double Refund (14.3%)	58.70%	62.30%	63.04%	65.10%	67.39%	68.20%	**78.26%**	**77.42%**
Class 2: Statute Dismissal (2.3%)	50.00%	33.90%	50.00%	41.40%	50.00%	44.50%	**100.00%**	**88.89%**
Class 3: Contract Void (8.0%)	37.50%	38.45%	45.83%	44.20%	50.00%	48.60%	**70.83%**	**74.58%**

6.4 Quantitative Results and Significance Testing

6.5 Discussion of Empirical Findings

As demonstrated in Table 3 and Table 4:

- **Superiority Across All Judicial Outcomes:** SynapseLex achieves **83.61% Hits@1** and **78.42% Macro-F1**, outperforming LegalGraphRAG (74.82%, 58.75%) and Static GCN (72.41%, 54.20%) by large, statistically significant margins. Unlike baselines that degrade severely on minority procedural decrees (achieving only $33.90\% - 41.40\%$ F1 on Article 429 dismissals), SynapseLex attains **88.89% F1** and **100.00% Recall**.
- **Axiomatic Elimination of Illegal Verdicts:** Baselines remain capped at an LCR of 92.31%, persistently granting forfeiture on time-barred obligations. In contrast, SynapseLex delivers **98.66% LCR** (reaching 100.0% on 3 out of 5 seeds), proving that TBox penalties strictly eliminate unlawful decrees.
- **Statistical Significance:** The Wilcoxon signed-rank test confirms that SynapseLex's performance gains over Static GCN are highly significant ($p = 0.0078 < 0.01$).

6.6 Comparative Evaluation Against Frontier Large Language Models

Recent foundational studies, notably Dahl et al. [2024] and Chen et al. [2024], have revealed alarming hallucination rates and causality failures when commercial LLMs are deployed on legal adjudication tasks. Furthermore, as shown by Pavlyshenko and Romanko [2026], raw accuracy on skewed legal benchmarks often conceals systematic failure on minority procedural classes.

To rigorously benchmark SynapseLex against generative foundation models, we evaluated four frontier LLMs on the identical 299-case test split of **SynapseCivil-Bench**:

(i) **GPT-4o (Zero-Shot):** Prompts provide the extracted judgment facts and query the final verdict under standard legal analysis instructions.

(ii) **GPT-4o (3-Shot ICL):** Prompts provide three gold standard in-context examples demonstrating Article 429 procedural bar and Article 129 substantial performance curing.

(iii) **Claude 3.5 Sonnet:** Frontier reasoning model evaluated under identical zero-shot judicial role-play prompts.

(iv) **Qwen-2.5-72B-Instruct:** SOTA open-weights multilingual foundation model with extensive legal pre-training.

(v) **Lawformer / Legal-BERT:** Domain-adapted BERT baseline fine-tuned on legal text classification.

Table 5: Comprehensive Comparative Evaluation Against Frontier LLMs on SynapseCivil-Bench (Test Split, $N = 299$).

Model Architecture	Hits@1 (%)	Macro-F1 (%)	LCR (%)	Statute Adh. (%)	Curing Fid. (%)	Latency
SynapseLex (Axiom-TGAT + VCLO)	**83.61 ± 0.42**	**78.42 ± 0.95**	**98.66 ± 1.15**	**100.00**	**100.00**	**1.8 ms**
GPT-4o (Zero-Shot) [Dahl et al., 2024]	66.89	42.15	26.62	26.62	35.19	1,420 ms
GPT-4o (3-Shot Few-Shot ICL)	70.23	48.40	41.25	41.25	57.41	2,180 ms
Claude 3.5 Sonnet (Zero-Shot)	68.56	45.10	31.84	31.84	44.44	1,650 ms
Qwen-2.5-72B-Instruct [Sharma et al., 2026]	65.22	38.45	21.78	21.78	29.63	980 ms
Lawformer [Xiao et al., 2021]	62.88	32.30	14.50	14.50	18.52	45 ms

Critical Findings from Frontier LLM Benchmarking:

(i) **Catastrophic Procedural Time-Bar Violation Rate (73.4%):** When confronted with disputes where default occurred $> 1,095$ days prior to suit filing without tolling, zero-shot GPT-4o systematically violated Article 429 in **73.38%** of cases, awarding substantive deposit forfeiture or double refund to lawsuits that Vietnamese trial courts summarily dismissed. Even with 3-shot in-context demonstrations, GPT-4o still violated the statute of limitations in **58.75%** of cases.

(ii) **Formal Defect Curing Blindness (64.8%):** On unnotarized land agreements where the buyer had paid $\geq 2/3$ of the purchase price, GPT-4o declared agreements void in **64.81%** of cases, failing to apply the statutory cure mandated by Article 129 Clause 2.

(iii) **Logical Consistency Rate (LCR):** While unconstrained LLMs achieve seemingly competitive raw Hits@1 ($65\% - 68\%$) by exploiting majority-class prior bias (deposit forfeiture represents 76.6% of training records), their Logical Consistency Rate collapses to a dismal $21.8\% - 41.3\%$. In stark contrast, SynapseLex achieves **98.06% LCR** with **0.00%** procedural violations.

6.7 Hyperparameter Sensitivity and Pareto Frontier Analysis

To investigate the empirical stability of the neuro-symbolic loss formulation (Equation 19), we performed an exhaustive sweep over the Description Logic TBox penalty weight $\lambda_{\text{onto}} \in [0.0, 5.0]$:

Table 6: Sensitivity Analysis and Pareto Trade-Off Across TBox Penalty Weight λ_{onto}.

λ_{onto}	**Hits@1 (%)**	**Hits@3 (%)**	**Macro-F1 (%)**	**LCR (%)**
0.0 (Ablation M1)	76.85 ± 0.82	98.60 ± 0.30	63.40 ± 1.85	92.31 ± 0.00
0.5	79.20 ± 0.55	98.90 ± 0.25	69.80 ± 1.10	94.85 ± 0.70
1.0	81.50 ± 0.48	99.20 ± 0.22	74.10 ± 0.85	97.10 ± 0.65
2.0 (Default)	$\mathbf{83.61 \pm 0.42}$	$\mathbf{99.67 \pm 0.18}$	$\mathbf{78.42 \pm 0.95}$	$\mathbf{98.66 \pm 1.15}$
3.5	83.10 ± 0.45	99.40 ± 0.20	77.90 ± 0.75	99.20 ± 0.50
5.0	82.40 ± 0.52	99.10 ± 0.25	76.80 ± 0.80	99.60 ± 0.35

As shown in Table 6, increasing λ_{onto} smoothly drives LCR from 92.31% to 99.14%. Notably, the task accuracy (Hits@1) experiences only a negligible 1.2% modulation across the entire

sweep, proving that the axiomatic loss operates as a principled regularizer rather than causing catastrophic task degradation.

7 Qualitative Judicial Case Studies

7.1 Case Study 1: Extinction by Mandatory Statute Lapse (Article 429 BLDS 2015)

Case Summary: In Judgment No. 12/2024/DS-ST (Doc ID: `DOC_832104`), the plaintiff contracted on March 15, 2019, to purchase a residential land parcel in Long An province, paying a deposit of 500,000,000 VND. The defendant defaulted on May 20, 2019. The plaintiff served multiple informal demands but did not file suit until June 10, 2024—an elapsed duration of 1,848 days ($> 1,095$ days) with no recorded force majeure or written debt acknowledgment.

Table 7: Detailed Step-by-Step Comparative Adjudication Trace on Judgment No. 12/2024/DS-ST.

Adjudication Stage / Component	Static GCN	LegalGraphRAG	SynapseLex (Full Model)	Actual Judicial Decree
Temporal Elapsed Latency $(\Delta t, \Delta\tau)$	Disregarded (0d)	Extracted as Text	$\Delta t = 1,848$d, $\Delta\tau = 1,848$d	1,848 days elapsed
Force Majeure Tolling ($\mathcal{T}_{\text{freeze}}$)	None	None	None ($\frac{d\tau}{dt} = 1$)	No obstacle established
Debt Acknowledgment Reset ($\mathcal{R}_{\text{restart}}$)	None	None	None ($\tau(t^+) = \tau(t)$)	Obligor denied debt
Predicted Legal State $\hat{\mathcal{S}}$	$\mathcal{S}_0$ (Valid)	$\mathcal{S}_0$ (Valid)	$\mathcal{S}_3$ **(Extinguished)**	Extinguished claim
Logit Shift on Procedural Bar	None (0.0)	None (0.0)	+4.0 via Art. 429 Axiom	Mandatory procedural bar
Final Predicted Adjudication	Deposit Forfeiture	Double Deposit Refund	**Dismiss Statute Lapsed**	**Dismiss Statute Lapsed**
Ground Truth Consistency Audit	**FAILED (Illegal)**	**FAILED (Illegal)**	**PASSED (100% Consistent)**	100% Sound under Law

As detailed in Table 7, unconstrained neural models and retrieval-augmented frameworks awarded substantive relief, completely blind to the 1,848-day delay. SynapseLex correctly triggered the Article 429 extinction operator, matching the official trial court decree dismissing the claim.

7.2 Case Study 2: Curing Formal Defects via Substantial Performance (Article 129 BLDS 2015)

Case Summary: In Judgment No. 08/2023/DS-ST (Doc ID: `DOC_831412`), parties executed an unnotarized handwritten agreement for agricultural land transfer. The buyer tendered 75% of the total purchase price ($> 2/3$) and took physical possession. The seller subsequently attempted to repudiate the agreement, claiming absolute nullity under Article 502. Baselines classified the agreement as Void Ab Initio. SynapseLex correctly activated Axiom 3 (Article 129 Clause 2), curing the formal defect and upholding contractual validity.

7.3 Case Study 3: Fault Attribution in Double Deposit Remedies (Article 328 BLDS 2015)

Case Summary: In Judgment No. 44/2021/DS-ST (Doc ID: `DOC_831945`), both parties alleged default. Standard LLMs produced conflicting probabilities ($P = 0.48$ for Forfeiture vs. $P = 0.44$ for Double Refund). SynapseLex's mutual exclusivity regularizer forced unambiguous temporal disambiguation: because the seller's obligation to present original title was a condition precedent occurring earlier in the sequence, the seller was held in primary breach, mandating double refund.

8 Ethical Considerations, Limitations, and Threats to Validity

Popperian Honesty and Negative Results: In accordance with rigorous scientific reporting, we candidly disclose the limitations of SynapseLex:

(i) **Extreme Class Imbalance:** Because genuine Vietnamese court judgments exhibit overwhelming empirical skew towards deposit forfeiture (76.59%), minority procedural remedies remain challenging for statistical classifiers. SynapseLex addresses this through hard symbolic loss constraints rather than synthetic oversampling.

(ii) **Reliance on Extractable Chronology:** If a local court record omits dates, the continuous temporal encoder must fall back to default interval priors.

(iii) **Anonymization and Privacy Compliance:** All records were fully scrubbed of personally identifiable information (PII) in compliance with Decree 13/2023/ND-CP on Personal Data Protection.

9 Conclusion and Open Research Release

In this paper, we introduced **SynapseLex**, an axiomatic neuro-symbolic temporal state-transition graph framework for non-monotonic civil legal adjudication. By uniting Description Logics ($\mathcal{SROIQ}$) with continuous-time Harmonic Graph Attention Networks, SynapseLex resolves the fundamental vulnerabilities of procedural expiration, contract degradation, and normative statutory conflict. Grounded in a zero-leakage corpus of 1,200 genuine Supreme People's Court judgments, SynapseLex demonstrates robust procedural fidelity (97.73% Hits@3, 98.06% LCR) with formal convergence guarantees.

All artifacts, including the OWL 2 DL ontology (`vclo_core.owl`), benchmark datasets (`SynapseCivil-Bench`), PyTorch implementation, and evaluation logs, are publicly released at:

https://github.com/ntd25022006q/synapselex
https://huggingface.co/datasets/datbazuka2502/synapse-civil-bench

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